Harsh Sinha

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Education

International Institute of Information Technology Bangalore **August 2023 – July 2028**
Integrated Master of Technology in Computer Science and Engineering ***[CGPA - 3.64/4.00]***

Experience

Centre for Technology Research and Innovations for Digital Governance **May 2025 – July 2025**
Summer Intern *IIIT Bangalore, Karnataka*

- Developed the Spring Boot backend for a Mess Management Application by extending RASP-generated CRUD code with custom APIs and implementing required business logic.
- Built user and admin routes and services, and implemented role-based security to control access for different roles.

Mentorship

Teaching Assistant **August 2025 – December 2025**
C Programming *IIIT Bangalore*

- Conducted C Programming labs, explained concepts, resolved doubts, debugged code, and evaluated assessments.

Projects

Food-e | *Node.js (Express), MongoDB, Redis, Cloudinary* **August 2025 – Present**

- Built the backend for a **college-wide mess management platform** serving **1,537 registered users** with $\sim$ **900 daily active users**, supporting daily and weekly menu access across the campus.
- Designed and implemented **role-based functionality** for **Users, Admins, Committee Members, and Caterer Managers**, covering ratings, feedback with images, sick-meal requests, announcements, and report generation.
- Implemented secure authentication using **JWT** (access and refresh tokens) along with **Microsoft OAuth**, restricting platform access to verified college users only.
- Improved performance and maintainability using **Redis caching, MVC architecture, Cloudinary** for file storage, and **cron jobs** for automated data cleanup.

AcadOS | *C, System Calls, File Locking, Linux* **May 2025**

- Built a concurrent **TCP client–server system** in C using **fork()**, with **role-based authentication** and **checks**.
- Implemented **file-based persistence** for users, courses, and enrollments, with **management** and **atomic updates**.
- Developed **menu-driven admin and student workflows** over sockets, including **course enrollment**.

EcoSynergy | *Node.js, Express, MongoDB, JWT, AI Integration* **April 2025**

- Built **backend APIs** for a **MERN platform** enabling creation and management of community sustainability initiatives with **role-based access**.
- Implemented **JWT authentication**, **authorization rules**, validation, **centralized error handling**, and **MongoDB (Mongoose) models**.
- Developed **collaborative post-event reporting** with document handling and **AI-powered summarization (Gemini API)**.

Achievements

Dean's Merit List | *IIIT Bangalore* **June 2024 – Present**

- Awarded by the Dean Academics for maintaining a CGPA of 3.6+/4.0 (9/10)

Technical Skills

Languages: Java, Python, C, C++, MySql, JavaScript, OCaml
Technologies/Frameworks: SpringBoot, Node.js, Express.js, Next.js, ReactJS
Databases: PostgreSQL, SQL, MongoDB, Redis

Leadership / Extracurricular

Food Committee, IIIT Bangalore | *Web Developer & Mentor* **August 2025 – Present**
∗ Developed and maintained the Food Committee platform **(Food-E)** and mentored juniors to support a live system.

Symphony — Music Club, IIIT Bangalore | *Guitarist & Core Member* **August 2023 – December 2025**
∗ Performed as a guitarist at club and college events and contributed to organizing inter-college music events.